

QA Practical Assessment: Scenario-Based Testing

Evaluation of End-to-End Test Design, Failure Modes & Edge Case Analysis

Candidate Name:	Date:
Position Title:	Total Time: 60 Minutes

INSTRUCTIONS FOR CANDIDATE

- Provide detailed, structured test scenarios for each scenario presented below.
- Categorize your test cases logically (e.g., *Functional, Boundary, Edge Cases, Security, UI/UX, Performance/Failure Recovery*).
- Focus on depth, realistic failure points, and backend/frontend synchronization—do not limit answers to happy path scenarios.

Question 1: E-commerce Checkout Flow Testing

SCENARIO DESCRIPTION

You are tasked with designing an end-to-end test suite for the checkout journey of an e-commerce platform. A customer adds a product to their shopping cart, proceeds through the checkout form, selects a payment method, and completes the transaction.

GIVEN REQUIREMENTS

- **Base Product Price:** ₹1,000
- **GST (18%):** ₹180
- **Shipping Charge:** ₹50
- **Total Payable Amount:** ₹1,230

TASK

Write a comprehensive test suite covering the entire checkout lifecycle from cart to order confirmation.

KEY AREAS EXPECTED IN YOUR RESPONSE

- Cart manipulation, item quantity changes, item removal, and inventory limit checks.
- Mathematical validation of tax rules, base price, shipping fees, discount codes, and final totals.
- Address input validation, mandatory field checks, PIN code handling, and contact details verification.
- Payment gateway integration, payment success/failure modes, user cancellations, session timeouts, rapid double-click handling, and network drops mid-transaction.
- Post-payment system state: order ID generation, payment status sync, email/SMS trigger verification, inventory decrement, and handling debited-but-failed order edge cases.

Question 2: Banking Application Login & Account Lock Mechanism

SCENARIO DESCRIPTION

You are evaluating the authentication security module of a high-security retail banking web application.

GIVEN REQUIREMENTS

- A user is permitted a **maximum of 3 consecutive incorrect password attempts**.
- Upon the **3rd failed attempt**, the system must **temporarily lock the account**.

TASK

Draft all necessary test scenarios to thoroughly validate this login and locking functionality.

KEY AREAS EXPECTED IN YOUR RESPONSE

- Standard credential verification, input formatting, whitespace handling, and case sensitivity checks.
- Boundary behavior surrounding the 1st, 2nd, 3rd (lock trigger), and 4th+ failed attempts.
- Account recovery mechanisms, lockout duration/timer accuracy, and post-lock login attempts with correct credentials.
- Multi-device/browser session behavior and active session termination upon locking.
- Security vulnerability checks: SQL injection/XSS inputs, password masking, automated brute-force resistance, and message feedback safety.

Question 3: File Upload Module Testing (HR Portal)

SCENARIO DESCRIPTION

You are testing the resume submission module on an enterprise HR recruitment portal.

GIVEN REQUIREMENTS

- **Allowed File Formats:** .pdf and .docx only.
- **Maximum File Size Limit:** 5 MB.

TASK

Detail all test scenarios required to validate this file upload feature end-to-end.

KEY AREAS EXPECTED IN YOUR RESPONSE

- Format validation (valid vs. prohibited extensions like .exe, .zip, .txt), corrupted files, and double extensions.
- Boundary value analysis for file sizes (below limit, exact 5 MB limit, above limit, and 0-byte empty files).
- Filename encoding, special characters, spaces, non-ASCII/Unicode strings, and duplicate filename handling.
- UI/UX behavior: drag-and-drop, progress bar accuracy, cancel/remove/replace actions, and user error messaging.
- Security controls: MIME-type/content verification, client-side vs. server-side validation, anti-virus scanning simulation, and prevention of executable payload uploads.